



Identification of potential FAK inhibitors using mol2vec molecular descriptor-based QSAR, molecular docking, ADMET study, and molecular dynamics simulation

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Abstract

This study aims to identify potential focal adhesion kinase (FAK) inhibitors through an integrated computational approach, combining mol2vec descriptor-based QSAR, molecular docking, ADMET study, and molecular dynamics simulation. A dataset of 437 compounds with known FAK inhibitory activities was used to develop QSAR models using machine learning algorithms combined with mol2vec descriptors. Subsequently, the most promising compounds were subjected to molecular docking against FAK to evaluate their binding affinities and key interactions. ADMET study and molecular dynamics simulation were also employed to investigate the pharmacokinetic, drug-like properties, and the stability of the protein–ligand complexes. The results showed that the mol2vec descriptor-based QSAR model established by support vector regression demonstrated good predictive performance ($R^2=0.813$, RMSE=0.453, MAE=0.263 in case of training set, and $R^2=0.729$, RMSE=0.635, MAE=0.477 in case of test set), indicating their reliability in identifying potent FAK inhibitors. Using this QSAR model and molecular docking, compound **21** (ZINC000004523722) was identified as the most potential compound, with predicted logIC₅₀ value and binding energy of 2.59 and −9.3 kcal/mol, respectively. The results of molecular dynamics simulation and ADMET study also further suggested its potential as a promising drug candidate. However, because our research was merely theoretical, additional in vitro and in vivo studies are required for the verification of these results.

Keywords Docking · FAK · mol2vec · Molecular dynamics · QSAR

Introduction

Focal Adhesion Kinase (FAK) is a member of the protein tyrosine kinase family and plays a crucial role in regulating cellular responses to various stimuli, such as integrins, cytokines, chemokines, and growth factors [1, 2]. Overexpression of FAK in cancer cells, which was observed in numerous cases, is believed to be involved in the development of tumors into a malignant phenotype [3, 4]. Recently, inhibiting FAK has emerged as a promising and potent strategy for the treatment of cancer, offering potential therapeutic benefits in combating this devastating disease [5].

The development of new drugs is a costly and time-consuming process. In order to save costs and enhance search efficiency, computer-aided drug design has become one of the important approaches in pharmaceutical research [6]. Among these techniques, the construction of Quantitative Structure–Activity Relationship (QSAR) models is one of the widely applied methods. QSAR models enable the prediction of the biological activity of compounds, even those not yet synthesized, thus saving time, costs, and aiding in synthetic planning. To date, several QSAR models predicting the inhibitory activity of Focal Adhesion Kinase (FAK) have been constructed, such as those by P. Cheng, et al. [7], F. Wang, et al. [8], S. Gosh, et al. [9], P. Kumar, et al. [10], X. Lu, et al. [11], P. Shirvani, et al. [12], and C. Sun, et al. [13]. However, the datasets used to build these models were relatively small, ranging from a few dozen compounds, which limited their applicability.

Natural Language Processing (NLP) is an important branch of Artificial Intelligence. Building upon the

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foundation of word2vec, a popular algorithm for NLP, in 2018, Sabrina Jaeger, et al. proposed the Mol2vec technique to generate vector representations for molecular structures [14]. Since then, this technique has been applied in various tasks related to chemical structure, including constructing QSAR/QSPR models and virtual screening. Some studies have demonstrated that models built using this type of descriptors yield comparable or even superior results to traditional molecular descriptor-based models [14, 15]. However, the application of the mol2vec technique for the virtual screening of active compounds is still limited, especially in the context of potential FAK inhibitory compounds. Consequently, our research aims to develop a Quantitative Structure–Activity Relationship (QSAR) model for predicting the FAK inhibitory activity of chemical substances using the mol2vec descriptors in combination with machine learning algorithms. Molecular docking, ADMET (Absorption, Distribution, Metabolism, Excretion, and Toxicity) studies, and molecular dynamics simulation were also carried out to further confirm the results obtained by the QSAR model.

Materials and methods

Dataset

The model-building dataset, including the molecular structures (represented as SMILES string) and their corresponding inhibitory activities against FAK (expressed as the half maximal inhibitory concentration, IC_{50} , in nM), was downloaded from the ChEMBL database (<https://www.ebi.ac.uk/chembl/>) [16]. However, because these compounds were collected from various published papers, it was crucial to curate data, especially the pharmacological activities. Therefore, only studies with FAK inhibitor assays conducted under the same conditions were selected to maintain data uniformity. The second dataset, which was used for virtual screening, was downloaded from the ZINC database and consisted of the structures of 224,205 compounds (Supplementary Information) [17].

Molecular descriptions calculation and preprocessing data

In this study, each molecule was represented as a vector of length 100, using a pre-trained mol2vec model according to the technique described by Sabrina Jaeger, et al. [14]. After that, the dataset was randomly divided into a training set (305 compounds, 70%) and a test set (132 compounds, 30%). The training set was used to build the QSAR model and for hyperparameter tuning. The test set, including substances that do not appear in the model training process, was used to evaluate the predictive ability of the obtained models.

In the next step, variables with a correlation coefficient greater than 0.9 were removed to ensure that multicollinearity was eliminated. For the dependent variable (IC_{50}), all these values were converted into the same unit as nM, then logarithmically obtained a new dependent variable, $\log IC_{50}$. After that, QSAR models representing the dependence of $\log IC_{50}$ on the obtained mol2vec molecular descriptors were established using machine learning algorithms.

Model building

In this study, seven algorithms were used to obtain QSAR models, including Random Forest (RF), Support vector regression (SVR), Linear Regression (MLR), Lasso Regression (LR), Ridge Regression (RR), XGBoost (XGB), and Artificial Neural Network (ANN). To avoid overfitting during model training, fivefold cross-validation was used instead of the entire training set. More specifically, the training set was divided into five parts. Four of them were used to build the model, and the rest for evaluation. In addition, for RF, SVR, XGB, and ANN models, grid search was used to select the optimal parameter to ensure the model had the best performance.

Model evaluation

To evaluate the obtained models, the coefficient of determination (R^2), mean absolute error (MAE), and root mean square error (RMSE) of the training set and test set were used. The MAE and RMSE were calculated according to the following formula:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - f_i)^2}$$

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - f_i|$$

where: y_i and f_i was the actual and predicted $\log IC_{50}$ value, respectively. A model is considered good if the coefficient of determination R^2 is close to 1 and MAE, RMSE is close to 0. As mentioned above, for the training set, instead of using R^2 , RMSE, and MAE of the entire training set, the fivefold R^2 , fivefold RMSE, and fivefold MAE were used to avoid overfitting.

In addition, to ensure that the obtained model is not caused by randomness, Y-randomization was performed by keeping the independent variable while shuffling the dependent variable value between observations, then rebuilding the model. The initial model was evaluated as a good model if the Y-randomization model had low quality, e.g., a low R^2 value.

Application domain

In order to get predicted results with an acceptable accuracy compared to the actual one, it is essential to define the application domain of the QSAR model. The application domain is the chemical space that defines the limitations and boundaries of the model, indicating where it can provide accurate predictions and where its results might be less trustworthy. The predicted values obtained by the QSAR model are reliable only if the compounds fall within the domain of application of the model. In this study, the application domain was determined using the five-nearest neighborhood approach. A compound was defined within the domain of application if:

$$D_i < D_k + Z \times S_k.$$

where D_k was the average of the Euclidean distance between each compound in the training set to its five nearest compounds; S_k was the standard deviation of D_k ; D_i was the average Euclidean distance between the compound to be predicted to the five closest compounds in the training set, Z was the parameter that determines the application domain size and was defined by the user. In other words, each QSAR model would have an appropriate value of Z [18, 19]. According to Tropsha, et al., the Z value is usually chosen as 0.5 [18]. However, in this study, four values of Z , including 0.25, 0.50, 0.75, and 1.00, respectively, were investigated to choose the most suitable application domain size.

Virtual screening

To virtual screen potential FAK inhibitors, the D_i value of each compound in the second dataset was calculated, then the compounds that fell within the application domain of the obtained QSAR model were selected. The corresponding $\log IC_{50}$ value of these substances was predicted by the above QSAR model. Compounds were identified as potential if they had a predicted IC_{50} value below 1 μM , which equals a $\log IC_{50}$ value below 3.

Molecular docking

To further screen for FAK inhibitors, molecular docking between the potential substances predicted by the QSAR model and focal adhesion kinase was performed to determine binding energy as well as the binding mode of these compounds with the enzyme. First, the structure of the protein (PDB ID: 2JKK) was downloaded from the RCSB protein data bank [20], followed by removing co-ligand, water molecules, adding hydrogen and assigning partial charges. After that, the ligand was converted to the corresponding 3D structure and saved as a pdbqt file using Autodock Tool 1.5.6 software.

After preparing the structure of the protein, and the ligands, molecular docking was carried out using AutoDock Vina [21]. Potential compounds were docked to the active site of the protein in a grid box with dimensions of $22.5 \times 22.5 \times 22.5$ ($\text{\AA}^3$). The position of the grid box was determined by 3 coordinates: center_x, center_y, and center_z, with the corresponding values of 8.497, 1.534, and 27.141, respectively. In this study, we used an energy

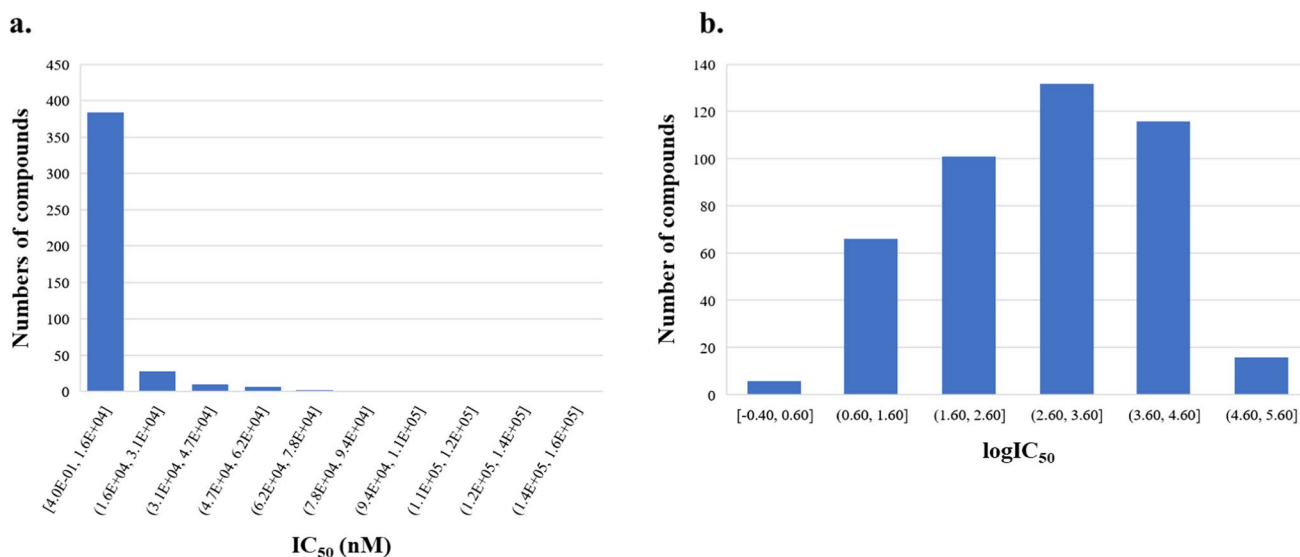


Fig. 1 Distribution histogram of (a) IC_{50} and (b) $\log IC_{50}$

range of 10 and an exhaustiveness of 8. Other parameters were set by default. The activity of these potential compounds was evaluated based on the binding energy. The lower the binding energy, the greater the binding affinity between the ligand and the receptor and vice versa. TAE-226, a potent FAK inhibitor that had been demonstrated in many studies, was used as the positive control [22–24]. The interactions between potential compounds and FAK were analyzed using Discovery Studio Visualizer software.

ADMET and drug-likeness study

Potential FAK inhibitors identified from the QSAR model and molecular docking were evaluated for their pharmacokinetics properties and drug-likeness, which was expressed through the number of principles of Lipinski's Rule of 5 that the compound violates, using the SwissADME server [25]. For toxicity, the in vitro mutagenicity (Ames test), skin sensitization, eye irritation and corrosion of these compounds were assessed using Toxtree 3.1 [26].

Fig. 2 Hyperparameter tuning results of SVR model

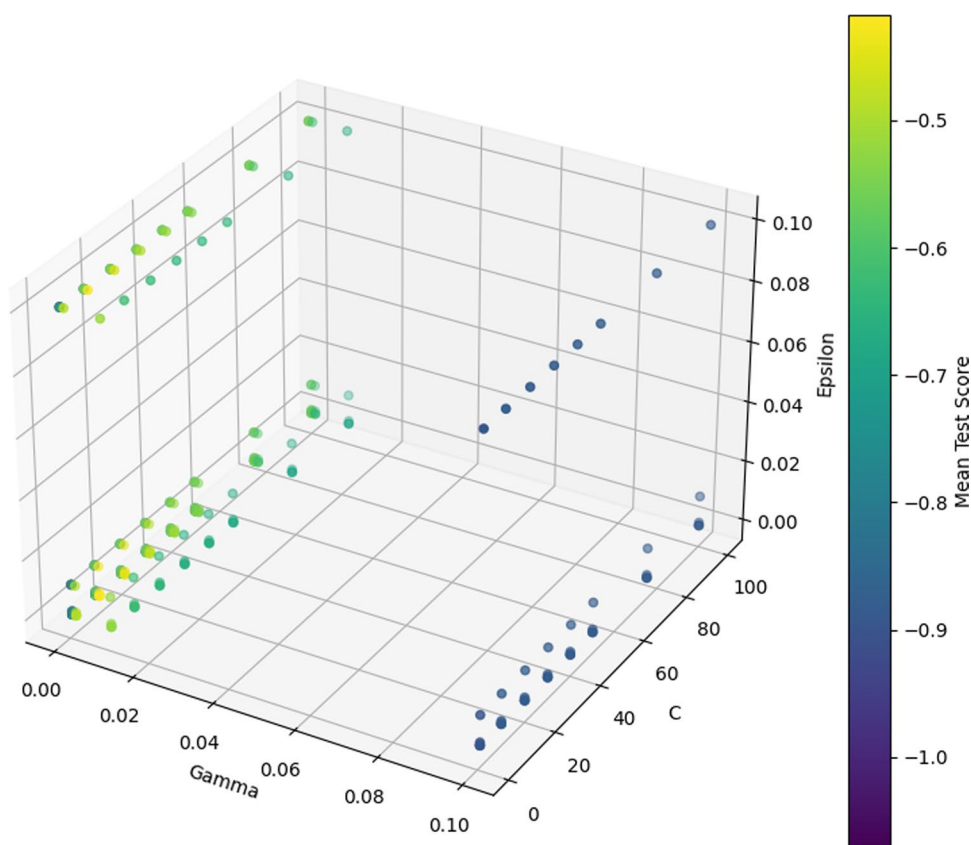


Table 1 The performance of established models on the training set and the test set

Model	Training set			Test set		
	R ²	RMSE	MAE	R ²	RMSE	MAE
RF	0.928	0.281	0.207	0.670	0.700	0.536
SVR	0.813	0.453	0.263	0.729	0.635	0.477
ANN	0.810	0.456	0.327	0.489	0.871	0.626
XGB	0.804	0.464	0.346	0.586	0.785	0.616
LR	0.630	0.638	0.489	0.552	0.816	0.625
RR	0.626	0.641	0.486	0.543	0.824	0.636
LaR	0.568	0.689	0.520	0.484	0.876	0.677

Molecular dynamics simulation

Based on the results of the QSAR, molecular docking and ADMET study, molecular dynamics was performed to evaluate the stability of the complex of FAK and the most potential substance using NAMD 2.14 software [27]. The simulation was carried out according to the protocol described in our previous study [28]. Briefly, the complex was firstly converted to a psf file using VMD 1.9.7 and CHARM-GUI server [29], then submerged in a water box with a padding size of 20 Å on each side of the complex ($96.4 \times 95.3 \times 92.6 \text{ Å}^3$), equivalent to 9250 TIP3 molecules of water. After neutralization with 0.15 M Na^+ and Cl^- and minimization for 1000 steps, the system was simulated for 50 ns under isothermal and isothermal conditions at 1 atm pressure and 37 °C (310 K). The results were analyzed using VMD 1.9.7 software, including the RMSD of the backbone, the RMSF of $\text{C}\alpha$, and the number of hydrogen bonds.

Results and discussion

QSAR model building

Preparation of the dataset

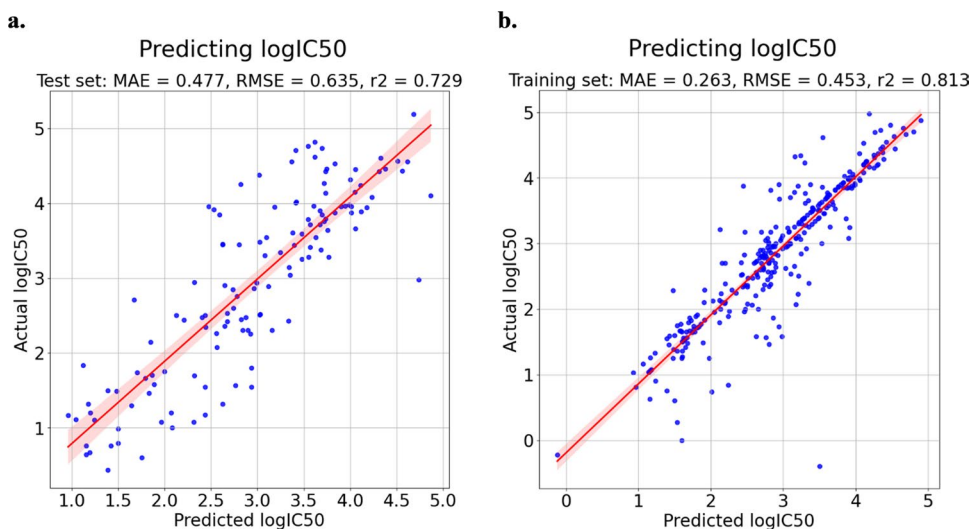
First, a dataset consisting of 2106 FAK inhibitors was collected from the ChEMBL database. However, because these compounds were assembled from a wide variety of sources and the protocol for evaluating FAK inhibitory effects was different in each paper, this phenomenon could significantly affect the quality of obtained QSAR models later. For example, in our collected dataset, the same compound TAE-226 (a FAK inhibitor used as a positive

control in many studies) was observed to have an IC_{50} value that ranges from 5.5 nM to 5200 nM. Therefore, not all 2106 compounds were selected for model building, but only those substances whose pharmacological activities were evaluated according to the same protocol were chosen. Although the results of biological assay might vary from person to person, we suppose our effort could minimize the heterogeneity of the data. The results obtained 437 compounds with IC_{50} values ranging from 0.4 nM to 156 μM . Since the IC_{50} values were positively skewed (Fig. 1a), the base-10 logarithm of the IC_{50} (denoted by $\log\text{IC}_{50}$) was used as the dependent factor instead of IC_{50} . The distribution of the dependent variable was shown in Fig. 1b.

Next, to represent each SMILES string as a vector, the pre-trained mol2vec model was used according to the technique described by Sabrina Jaeger, et al. [14]. The results obtained 100 molecular descriptors corresponding to each compound. In the next step, the dataset was randomly divided into two sets: the training set including 305 compounds (accounting for 70%), and the test set of 132 compounds (accounting for 30%). The training set was used for QSAR model building using different algorithms and the test set was utilized to evaluate the predictive ability of the obtained models.

Although previous studies had used all these molecular descriptors for model building, the existence of multicollinearity among these 100 variables was found in this study. Based on our initial investigations, removing multicollinearity could significantly increase the quality of obtained QSAR model. Although a reduction of the number of independent variables would make it difficult for the model interpretation, we believed that the removal of multicollinearity in order to increase the quality of the model was acceptable for this study since the aim of our study was to focus on

Fig. 3 Actual $\log\text{IC}_{50}$ graph and predicted value obtained from SVR model of (a) Test set and (b) Training set



the prediction of $\log IC_{50}$ value of screened potential compounds. Finally, each substance in the dataset was described by 49 molecular descriptors.

Model building and evaluation

In this study, seven algorithms, including Random Forest (RF), Support vector regression (SVR), Linear Regression (MLR), Lasso Regression (LR), Ridge Regression (RR), XGBoost (XGB), and Artificial Neural Network (ANN), were used to establish seven corresponding QSAR models representing the dependence of FAK inhibitory activity on the molecular descriptors. For RF, SVR, ANN, and XGB models, the grid search was used to select optimal parameters for the models. The investigated parameters included the number of decision trees ($n_{estimators}$), the maximum depth allowed (max_depth), the minimum number of samples required to split an internal node ($min_samples_split$), and the minimum number of samples required to be at a leaf node in a decision tree ($min_samples_leaf$) for RF model; C , γ and ϵ for the SVR model; the number of hidden layers, the number of neurons in the hidden layer, α and learning rate for ANN models; learning rate, $n_{estimators}$ and max_depth for the XGB model. To avoid overfitting, fivefold cross-validation was utilized, instead of using the entire training set.

The grid search results for the SVR model were shown in Fig. 2. As could be seen from Fig. 2, the C parameter, also known as the cost parameter, which controls the trade-off between the complexity and performance of the SVR model, had no effect on the model quality. Therefore, the smallest value of C as $C = 10$ was chosen. Regarding the remaining two parameters, ϵ and γ , from the analysis results, the optimal values of these two parameters in the investigated range were 0.1 and 0.001, respectively. The results of hyperparameter tuning for the remaining models were presented in the Supplementary Information.

Based on the grid search results, the best models corresponding to each algorithm: random forest, support vector regression, artificial neural network, XGBoost, linear regression, ridge regression, and Lasso regression were built. The performance of these models on the training set and test set was presented in Table 1.

From the results of Table 1, it showed that the RF model had the best performance on the training set with an R^2 value of 0.928 and RMSE and MAE of 0.281 and 0.207, respectively, which was followed by SVR, ANN, and XGB models with an R^2 values ranging from 0.804 to 0.813, RMSE from 0.453 to 0.464 and MAE from 0.263 to 0.346. Then there were two models, LR and RR, with R^2 above 0.6, and the worst was the LaR model, with R^2 less than 0.6. However, the performance of these models on the test set was a little bit different from the results on the training set. Specifically, although the RF model performed very well on the training set, this model showed an average predictive ability on the test set with R^2 of 0.670 and RMSE of 0.700. Similar results were also observed in the case of two models, ANN and XGB. Among the built models, only the SVR model showed acceptable predictive ability on the test set with the values of R^2 , RMSE, and MAE of the training set and the test set not differing too much. The graph illustrating the actual $\log IC_{50}$ and the value predicted by the SVR model showed that there was a high correlation between the predicted and actual values (Fig. 3). In other words, this model can be used to predict the $\log IC_{50}$ value from the molecular descriptors.

To evaluate the effectiveness of mol2vec descriptions in building QSAR models, we also established another model using support vector regression and traditional molecular descriptors calculated by PaDEL software as a baseline model and compared the performance of the two models.

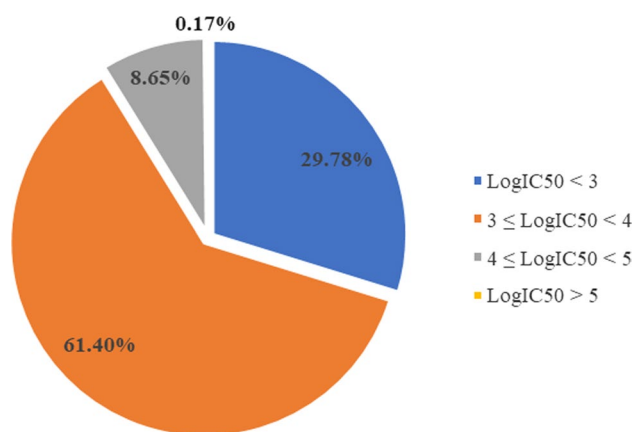


Fig. 4 Results of virtual screening using QSAR model

Table 2 Performance of the QSAR on the test set with various application domain size

Model	D_k	S_k	Z	Inside AD				Outside AD			
				No	R^2	RMSE	MAE	No	R^2	RMSE	MAE
SVR	6.84	6.88	0.25	70	0.737	0.633	0.461	62	0.711	0.637	0.495
			0.50	91	0.736	0.634	0.471	41	0.694	0.637	0.490
			0.75	104	0.740	0.624	0.464	28	0.680	0.673	0.527
			1.00	109	0.728	0.633	0.471	23	0.732	0.644	0.505

Specifically, the substances in the dataset were converted from SMILES strings to 3D structures, then passed through PaDEL software to obtain 1444 molecular descriptors. Next, the molecular descriptors with constant values, and those having a correlation coefficient with other descriptors of more than 0.9 were removed. After that, the baseline model was obtained using machine learning algorithms combining with hyperparameter tuning following the same protocol as for the model based on mol2vec descriptors. The results showed that although this baseline model performed relatively well on the training set with R^2 , RMSE, and MAE of 0.866, 0.383, and 0.164, respectively, the performance of this model on the test set was not as expected with R^2 , RMSE, and MAE of 0.415, 0.932, and 0.737, respectively. This result suggested that the approach to building QSAR models based on mol2vec descriptors used in this study was more appropriate.

Y-randomization

Y-randomization is a technique used to assess the significance of a model by randomizing the response variable ($\log IC_{50}$) while keeping the independent variables unchanged. By comparing the performance of the randomized model with the original model, one can make sure whether the quality of the obtained model is caused by chance. In this study, Y-randomization was conducted 100 times with the evaluation parameter of $R^2_{5\text{-fold}}$. The lower the R^2 value of these random models, the more stable the original model was, and vice versa. The results of the analysis of 100 random models of Y-randomization revealed that the original model outperformed the random models significantly, with an initial model's R^2 of 0.813 and an average of Y-randomized model's R^2 of 0.325. In other words, the obtained QSAR model had predictive power and was not merely fitting noise in the data.

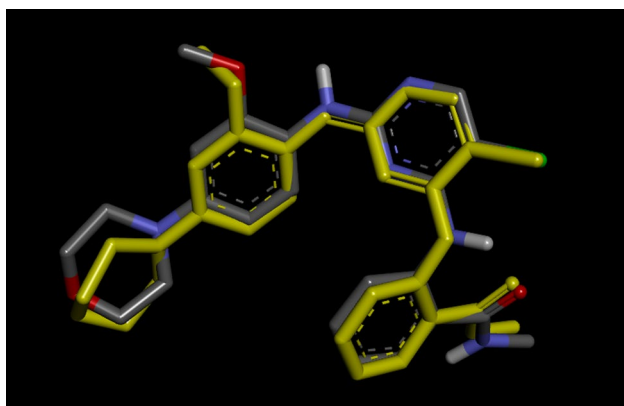


Fig. 5 Comparison of the redocked binding pose (yellow) with the experimental pose of the co-ligand

Application domain

Each QSAR model has a specific application domain based on the chemical structure characteristics of the compounds used to build the model. For compounds which structure is too different from the one in the dataset, which is considered as outside the application domain of the model, the prediction results of the QSAR model on this substance will become less accurate and unreliable [30]. Therefore, in order to apply the QSAR model to screen potential compounds with FAK inhibitory effects, it is necessary to determine the application domain of the model. Currently, to determine the application domain of a QSAR model, many techniques such as range-based, distance-based, and density-based methods can be used [19]. In this study, we chose the method based on Euclidian distance to determine the application domain of the best QSAR model obtained above.

For the SVR model, to choose the appropriate Z parameter value, we investigate four values, including 0.25, 0.50, 0.75, and 1.00. The corresponding results, consisting of the number of compounds in the test set that were inside and outside the AD as well as the performance of the model on these compounds were shown in Table 2. In the case of $Z=0.25$, up to 62 substances in the test set were outside the application domain of the model. At the same time, the predictive ability of the model on compounds in and out of AD was almost equivalent. This result indicated that the application domain, in this case, was so small that many compounds, although the prediction results were accurate, were not in the AD of the model. In the case of $Z=0.50$, the number of substances inside AD increased, and the performance of the model on these compounds also showed a significant difference compared to substances outside AD. These results suggested that AD with $Z=0.50$ is more suitable than $Z=0.25$. Similarly, for $Z=0.75$, the predictive power on compounds inside AD remained unchanged, but on those outside AD continued to trend down. However, when $Z=1.00$, the performance of the model on substances inside the application domain decreased. These results suggested that the AD of the model was larger than necessary, causing some compounds that were supposed to be outside AD to enter, thereby reducing the predictive ability of the model. Based on these results, the value $Z=0.75$ was chosen to calculate the application domain of the model. Compounds with a distance to the five nearest compounds in the training set less than the threshold value $D_k + 0.75 \times S_k = 12.00$ were considered in the application domain of the model and continue to be screened by the built QSAR model.

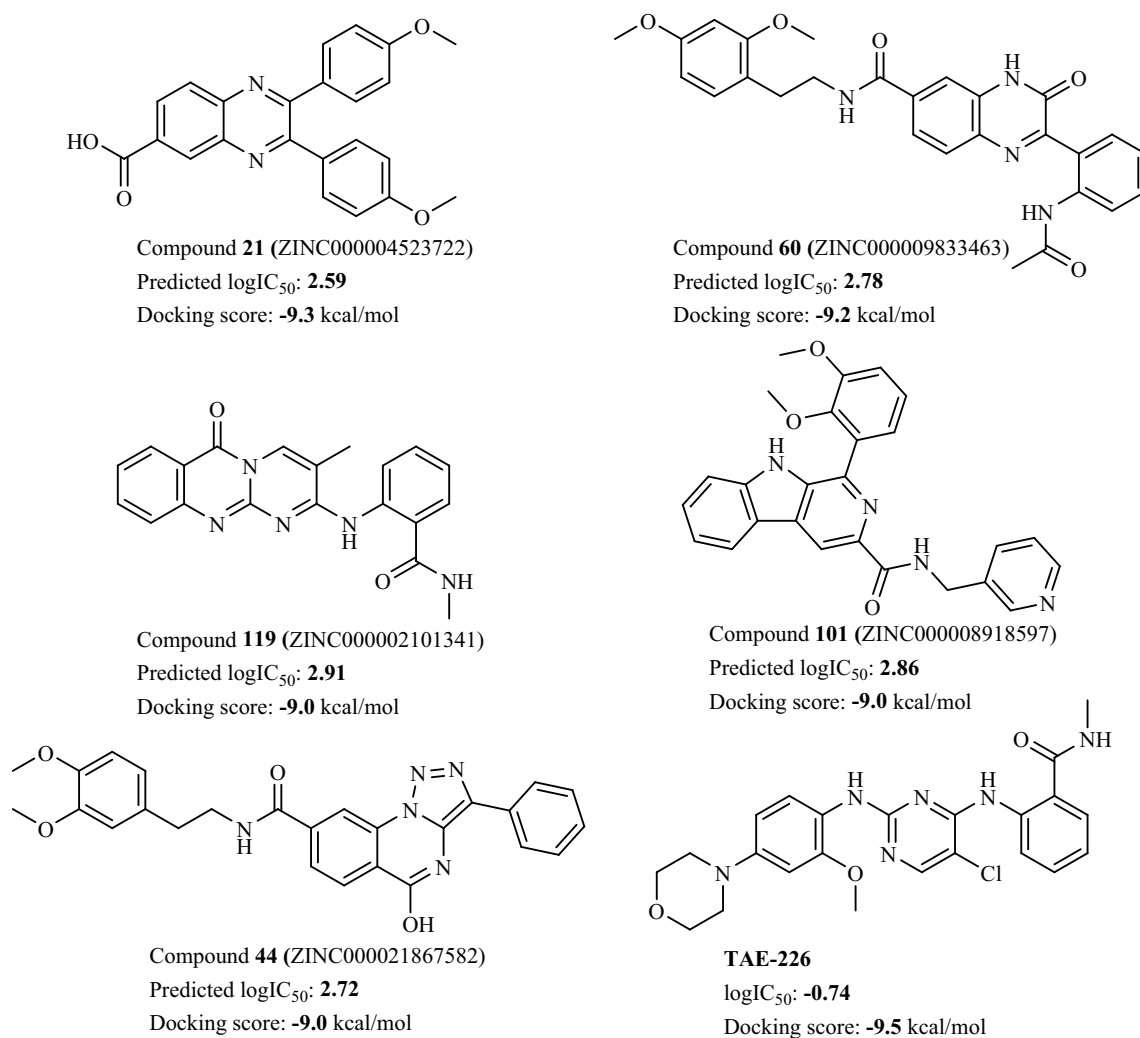


Fig. 6 Five potential FAK inhibitors identified from molecular docking results and the positive control

Screening for potential FAK inhibitor compounds using the QSAR model

In order to identify potential compounds with FAK inhibitory effects, first, the D_i values of 224,205 compounds in the dataset were determined. A substance was considered in the application domain of the model if the D_i value satisfies the condition $D_i \leq 12.00$. Then the FAK inhibitory effect of these compounds was assessed through the logIC₅₀ value that would be predicted by the QSAR model built above. The results showed that, among the compounds in the application domain of the SVR model, 8.8% of them had logIC₅₀ values greater than 4, and more than 60% of the compounds had logIC₅₀ values in the 3–4 range. The number of substances with logIC₅₀ less than 3 was 179 compounds, accounting for nearly 30% (Fig. 4). In this study, we chose a logIC₅₀ value of 3 corresponding to an IC₅₀ value below 1 μ M as the threshold. In other words, compounds with a logIC₅₀

value less than 3 would be selected for further screening by molecular docking technique.

Molecular docking

To further screen for potential FAK inhibitors, molecular docking was used then promising compounds were selected based on the binding energy with the protein (FAK). Before carrying out molecular docking between FAK and 179 potential compounds identified from section “[Screening for potential FAK inhibitor compounds using the QSAR model](#)”, our protocol was validated by redocking co-ligand (TAE-226) and comparing it with experimental results via RMSD value. The method was considered reliable if the RMSD value was less than 2 Å [31]. The results showed that our protocol had high accuracy with the RMSD value between the redocked and the actual conformations of

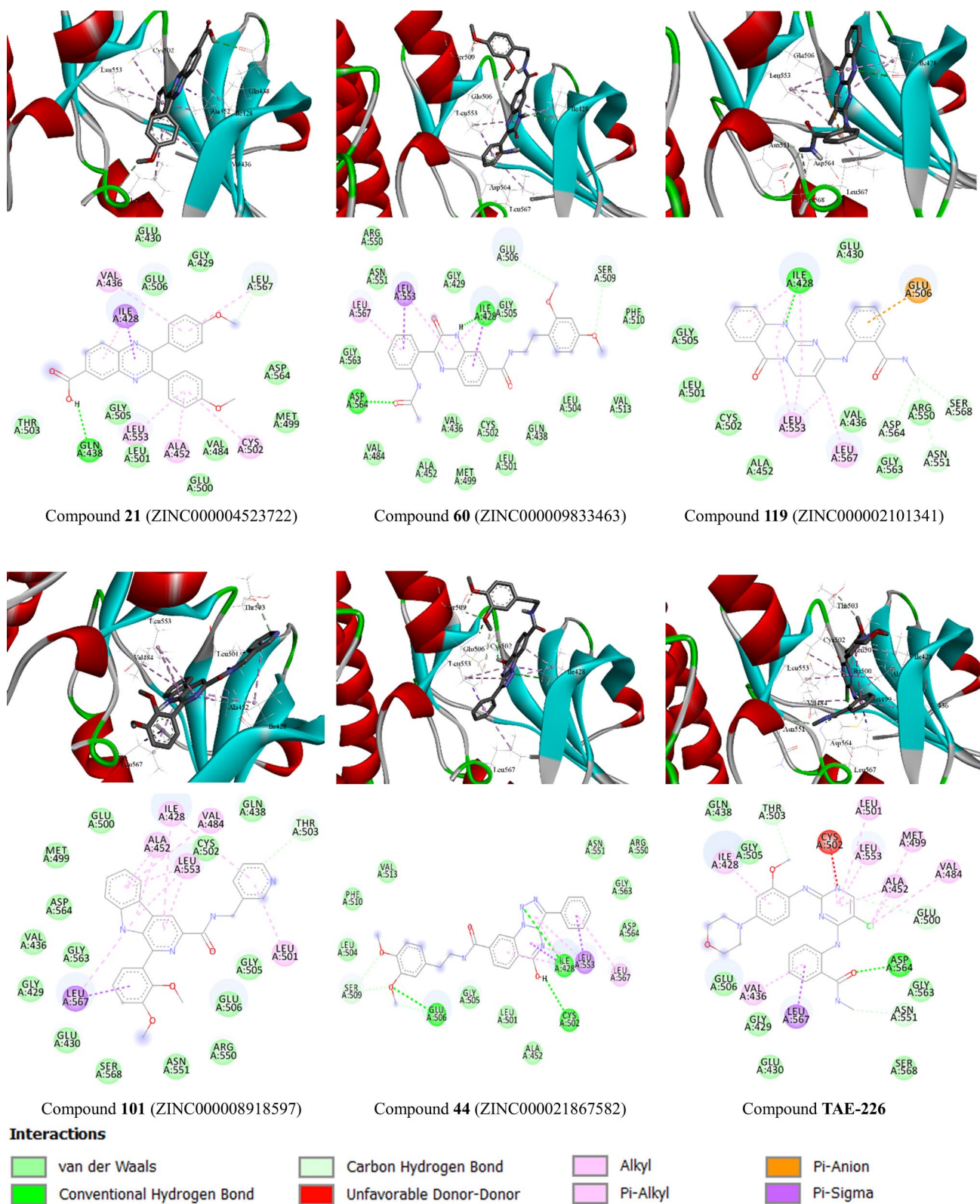


Fig. 7 The binding mode of 5 potential compounds with FAK

Table 3 ADMET and drug-likeness study of 5 potential compounds

Properties	Compound 21	Compound 44	Compound 60	Compound 101	Compound 119
GI absorption	High	High	High	High	High
BBB permeant	No	No	No	No	No
Pgp substrate	No	Yes	Yes	Yes	No
CYP inhibitor	CYP1A2, CYP2C19, CYP2C9, CYP2D6, CYP3A4	CYP1A2, CYP2C19, CYP2C9, CYP2D6, CYP3A4	CYP1A2, CYP2C19, CYP2C9, CYP2D6, CYP3A4	CYP1A2, CYP2C19, CYP2C9, CYP2D6, CYP3A4	CYP1A2, CYP2C9
Bioavailability Score	0.56	0.55	0.55	0.55	0.55
Lipinski #violations	0	0	0	0	0
Leadlikeness #violations	2	3	2	2	1
Synthetic Accessibility	2.80	3.42	3.57	3.37	2.81
In vitro mutagenicity	No	Yes	Yes	Yes	Yes
Skin sensitization	No	No	Acyl Transfer agent	Acyl Transfer agent	Acyl Transfer agent
Eye irritation and corrosion	No	No	No	No	No

0.5177 Å (Fig. 5). Therefore, the molecular docking technique in this study could be used to screen for potential FAK inhibitors.

The docking results of 179 potential compounds identified by the QSAR model and FAK showed that the binding affinity of these compounds to the respective enzyme ranged from -7.0 kcal/mol to -9.3 kcal/mol with an average value of -8.3 kcal/mol (Supplementary Information). According to S. Long, et al. [32] and S. A. Khan, et al. [33], binding energies less than -4.25 kJ/mol indicated that the substrate might interact with the protein while the binding affinity below -5.0 kJ/mol showed good ligand–protein interactions. Binding energies less than -7.0 kJ/mol indicated that the substrate could strongly interact with the enzyme. Thus, all 179 compounds could bind well to FAK with an average binding affinity of -8.3 kcal/mol. This result was also consistent with the outcome obtained by the QSAR model when these compounds were all identified as potential FAK inhibitors with the predicted IC_{50} value of less than 1 μ M. In this study, TAE-226, which had an experimental IC_{50} value of 5.5 nM, equivalent to a $\log IC_{50}$ of -0.74 , was used as a positive control. The docking results revealed that the binding energy value of TAE-226 and FAK was determined as -9.5 kcal/mol. Comparison with the binding energies of 179 potential compounds showed that none had a lower binding affinity than the positive control which was also consistent with the prediction results of the QSAR model when the predicted $\log IC_{50}$ values of these 179 compounds were between 2.00 and 3.00 . Although none of these compounds exhibited greater interactions with FAK than TAE-226, the pretty low protein binding energy values of these compounds also suggested their potential in the development of novel FAK inhibitors. Therefore, compounds with binding affinity values less than or equal to -9.0 kcal/mol were chosen to

carry out further studies. The structures of these compounds were shown in Fig. 6.

The binding mode of five potential compounds that had docking scores less than or equal to -9.0 kcal/mol were shown in Fig. 7. The results revealed that the positive control TAE-226 could strongly interact with FAK by forming hydrogen bonds with Cys502 and Asp564 residues. Besides, we also observed Van der Waals interactions, π -alkyl, and π - σ interactions between TAE-226 and FAK. The interaction analysis between compounds **21**, **44**, **60**, **101**, and **119** also gave similar results. Based on the docking results, important residues related to the biological effects of FAK were suggested, including Ile428, Val436, Gln438, Ala452, Cys502, Leu553, and Leu567.

Previous research had described the diverse landscape of FAK inhibitors, including various scaffolds such as diphenylpyrimidines, 1,2,4-triazines, 7H-pyrrolo[2,3-d]pyrimidines, thieno[3,2-d]pyrimidines, 1,3,4-oxadiazoles, 1,2,4-triazole, benzotriazole, arylquinolines, and thiosemicarbazones [34]. For instance, TAE-226, a well-known FAK inhibitor, is classified under diphenylpyrimidines derivatives. However, our potential compounds (compound **21**, **44**, **60**, **101**, and **119**) exhibit distinct structural motifs that deviate from the commonly observed scaffolds in FAK inhibitors (Fig. 6). Specifically, compounds **21** and **60** feature a quinoxaline framework, while compounds **44** and **119** are quinazoline derivatives. Additionally, the inclusion of 9H-pyrido[3,4-b]indole in compound **101** introduces a novel moiety that may enhance binding affinity and selectivity toward FAK.

ADMET and drug-likeness study

One of the important factors determining whether a drug could be used clinically or not is the pharmacokinetic and

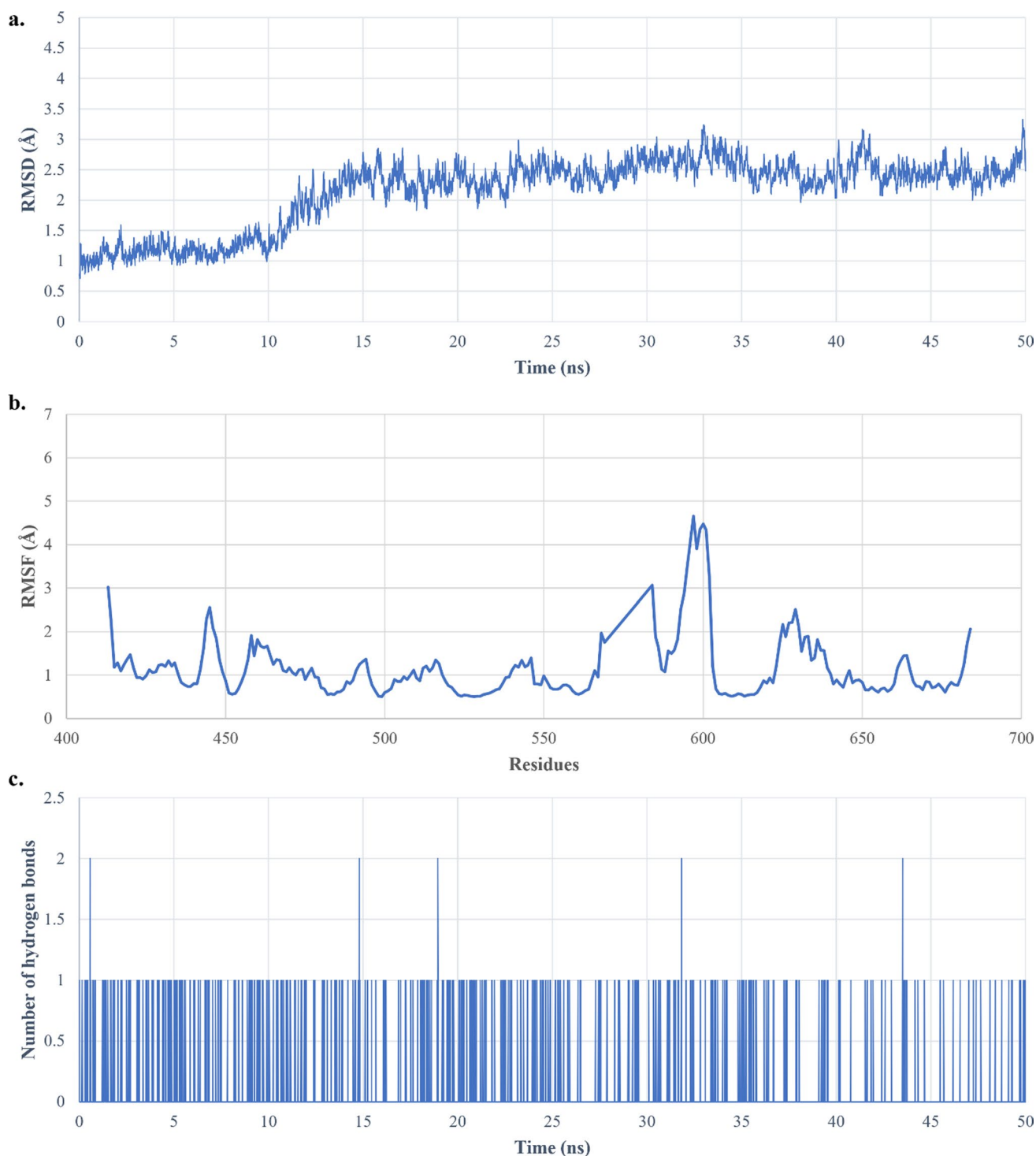


Fig. 8 RMSD of the backbone (a), RMSF of Cα (b), and the number of hydrogen bond (c) during the 50 ns simulation

toxicological properties (absorption, distribution, metabolism, excretion, and toxicity—ADMET) of that drug. According to Helen Dowden, et al. [35] and Richard K. Harrison [36], the failure rate of research and development of new drugs in phase I/II was as high as 90%, of

which 40–45% of cases were related to problems of toxicity and drug-like properties. Therefore, the development of compounds with suitable ADMET properties is extremely important and necessary. In this study, the pharmacokinetic properties, including absorption, distribution, metabolism,

and excretion, as well as drug-like properties of five potential compounds (compounds **21**, **44**, **60**, **101**, and **119**) were predicted and evaluated using the SwissADME server. The toxicity of these compounds, including in vitro mutagenicity (Ames test), skin sensitization, and Eye irritation and corrosion was studied using Toxtree 3.1 software. The results were presented in Table 3.

From the results of Table 3, it was shown that all five compounds had good absorption through the gastrointestinal tract with high bioavailability. In addition, none of the compounds cross the blood–brain barrier. Of the five substances **21**, **44**, **60**, **101**, and **119**, only two compounds were not Pgp substrates, including compounds **21** and **119**. At the same time, none of these five substances violate any principle of Lipinski's Rule of Five. In other words, these compounds all had suitable drug-like properties. However, the results of toxicity prediction showed that, except for compound **21**, the remaining compounds, including compounds **44**, **60**, **101**, and **119** were mutagenic agents in in vitro model. In addition, compounds **60**, **101**, and **119** also have the potential to cause skin toxicity due to interaction with the enzyme acyl transferase. For this reason, out of the five potential compounds identified by the QSAR model combined with molecular docking, only compound **21** was selected as a potential lead for further researches.

Molecular dynamics simulation

To evaluate the stability of the complex between FAK and compound **21**, molecular dynamics simulation was carried out for 50 ns, using NAMD software. The results were analyzed using VMD 1.9.7 software, including the RMSD values of the backbone, the RMSF of C α and the number of hydrogen bonds during 50 ns simulation (Fig. 8). The results showed that for the RMSD value, the complex reached equilibrium after about 15 ns and maintained until the end of the simulation process, with the average RMSD value calculated from 15 to 50 ns being 2.46 Å. The highest RMSD value in the 50 ns simulation observed was 3.33 Å. These results indicated that the FAK – compound **21** complex was stable, with little deviation before and after 50 ns of simulation. In other words, compound **21** was capable of stably binding to FAK.

Besides RMSD, the RMSF of C α was also determined. The results showed that most residues were stable during the simulation time with the corresponding RMSF value less than 3 Å. Although some fluctuations in the residues Arg597 (4.65 Å), Thr600 (4.48 Å), Phe599 (4.36 Å), Ser601 (4.34 Å), Phe596 (4.05 Å), Arg598 (3.91 Å) were observed, these residues were not important residues related to the FAK inhibitory effect of compound **21** as identified by molecular docking (Ile428, Val436, Gln438, Ala452,

Cys502, Leu553, and Leu567). These results suggested that compound **21** was capable of sustained stable interaction with FAK. In the case of the number of hydrogen bonds, the analysis also showed that during the 50 ns simulation, compound **21** could form with FAK residues from 1 to 2 hydrogen bonds. The average number of hydrogen bonds in the simulation time was determined to be 0.08 ± 0.28 bonds. This result suggested that hydrogen bonding might be just one of the interactions rather than the determining factor for the strong affinity of compound **21** toward FAK.

Conclusion

In this study, using support vector regression, a QSAR model representing the relationship between the focal adhesion kinase (FAK) inhibitory activities and mol2vec molecular descriptors of 437 substances was obtained. By utilization of this QSAR model and molecular docking, from a library of 224,205 substances, compound **21** (ZINC000004523722) was identified as the most potential compound, with predicted logIC₅₀ value and binding energy of 2.59 and – 9.3 kcal/mol, respectively. The results from the ADMET study and molecular dynamics simulation further suggested that compound **21** was a potential lead for the development of novel FAK inhibitor drugs. However, because our research was merely theoretical, additional in vitro and in vivo studies are required for the verification of these promising results.

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Declarations

Competing interests The authors declare no competing interests.

Conflict of interest The authors declare no competing financial interest.

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